

EXAMPLE 9.19

Problems for Section 9.2: Analysis by Linearization

9.1 Fig. 9.56 shows a simple pendulum system in which a cord is wrapped around a fixed cylinder. The motion of the system that results is described by the differential equation

$$(l + R\theta)\ddot{\theta} + g \sin \theta + R\dot{\theta}^2 = 0,$$

where

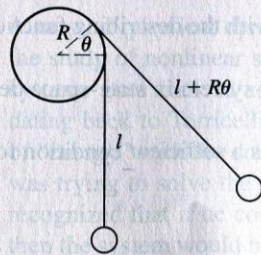
l = length of the cord in the vertical (down) position,

R = radius of the cylinder.

(a) Write the state-variable equations for this system.

Figure 9.56

Motion of cord wrapped around a fixed cylinder



(b) Linearize the equation around the point $\theta = 0$, and show that for small values of θ , the system equation reduces to an equation for a simple pendulum—that is,

$$\ddot{\theta} + (g/l)\theta = 0.$$

9.2 The circuit shown in Fig. 9.57 has a nonlinear conductance G such that $i_G = g(v_G)$ = $v_G(v_G - 1)(v_G - 4)$. The state differential equations are

$$\frac{di}{dt} = -i + v,$$

$$\frac{dv}{dt} = -i + g(u - v),$$

where i and v are the state variables and u is the input.

(a) One equilibrium state occurs when $u = 1$, yielding $i_1 = v_1 = 0$. Find the other two pairs of v and i that will produce equilibrium.

(b) Find the linearized model of the system about the equilibrium point $u = 1$, $i_1 = v_1 = 0$.

(c) Find the linearized models about the other two equilibrium points.

